Department of Information Technology
Faculty of Computing
Sri Lanka Institute of Information Technology

IT2140 : Database Design and Development

Year 2 Semester 1

2026

Assignment Title	Enhanced Entity Relationship Diagram
Learning outcomes covered	LO 1: Design conceptual and logical database schema for a data intensive application
Assignment Mode	Group
Maximum Marks	100
Contribution to the Final Grade	5%
Date published	20th August 2026
Deadline for submissions	30th August 2026
Mode of Submission	Courseweb Link submission

Description of the Assignment

Each group has already been assigned a system topic. Your assignment consists of two main parts:

Part A - Requirement Gathering – Identify requirements for your assigned system.

Part B - EER Diagram Design – Create an Enhanced ER diagram based on the requirements you gathered.

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Part A – Requirement Gathering (40%)

Instructions

1. Gather the requirements for your assigned system by conducting appropriate requirements gathering activities.
2. Document the following in a **Requirement Specification Document (2–3 pages)**.
    - **System Purpose** – overall aim of the system.
    - **Users** – who will use the system.
    - **Functional Requirements** – main features and operations.
    - **Non-Functional Requirements** – e.g., security, scalability, performance.
    - **Constraints/Assumptions** – any limitations or conditions.
3. Include screenshots of the requirement-gathering process. Embed these screenshots in your document as evidence of the requirement gathering activity. (**No page limit**.)

📌 Deliverable:

- GroupID_Assignment01_Requirements.pdf

Part B – EER Diagram Design (60%)

Instructions

1. Based on the requirements gathered in Part A, design an **Enhanced ER (EER) Diagram**.
2. Your diagram must include:
    - Entities and attributes (simple, composite, multivalued if applicable).
    - Relationships with correct cardinalities (1:1, 1:N, M:N).
    - Specialization/Generalization (ISA hierarchies).
    - Recursive relationships (if applicable).
    - Aggregate relationships (if applicable)

3. Clearly indicate **primary keys** and **participation constraints**.
4. You may use software tools such as **Draw.io, Lucidchart, ERDPlus, or any ER modeling tool**.

📌 Deliverable:

- GroupID_Assignment01_EER.pdf

Final Submission

Submit **two separate PDF files**:

1. Requirement Specification Document
2. EER Diagram

📌 File naming convention:

- GroupID_Assignment01_Requirements.pdf
- GroupID_Assignment01_EER.pdf

📌 Deadline: *30th August 2026*
